Igor Bryzhikov

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About Me

MIPT SAMCS student with a strong foundation in mathematics, physics, and software engineering. Experienced in end-to-end ML development: from task formalization and data preparation to model validation and deployment. Interested in agentic systems and LLM-based pipelines. Looking to apply CS and analytical skills in an industrial R&D environment.

Education

MIPT, SAMCS Moscow
Bachelor, "Natural and Computer Sciences", 2025 – present
09.03.01 "Informatics and Computer Engineering" **GPA: 8.7**

Experience

T-Bank July 2026 – October 2026
BI Analyst (internship)
Data Security team

- The team finds and masks sensitive data in the bank's DWH (Data Warehouse); I automated its processes and built analytics for them.
- **Incident auto-closure** (full cycle: requirements → analysis → design → production). Incidents on deleted tables used to be closed by hand; automated the process and **took part of the operational load off the team**. Designed a mechanism that rules out **over 90% of wrongful closures**.
- **Re-scan queue:** checks of tables for sensitive data lagged behind the growth in their number. Optimised the object selection rules around the operational load on on-call engineers and **ensured the quarterly target was met**.
- **Effectiveness metrics:** the effect of the automated pipeline that routes sensitive-data detection alerts to data-owner teams was not measured. Gathered requirements, designed and computed the metric; showed a **24% reduction in operational load**.
- **AI task review:** built an LLM skill that automatically checks Jira tasks before human review; **the interns' mentor's review load dropped 2.5 times**.
- **AI access assistant:** built an LLM skill that shows a user which fields of a table are hidden as sensitive data; average time to filing an access request fell **from 9 to 2.75 minutes**.
- **Masking:** some tables hid sensitive data in a non-standard way that would stop working after a change of the access-control system. Organised the move to the standard mechanism together with data-owner teams; **eliminated the risk of losses from a sensitive-data leak worth over 10 million RUB**.

Skills

Backend/Data: Python, C++, SQL, SQLAlchemy, FastAPI, PyTorch, REST API.

ML/NLP: LLM, prompt engineering, RAG architectures, LoRA, evaluation, feature engineering.

Algorithms & Data Structures: sorting, trees, hash tables, graphs, dynamic programming.

MLOps / Infra: Git, Linux, Bash, Docker, CI/CD, Apache Superset (T-Bank Proteus), Notion, GitHub Issues.

Languages: Russian (native), English (B2).

Achievements

- **Interregional Mathematics Olympiad for Departmental Educational Organizations**
  - Prize-winner of the final stage
- **All-Russian School Olympiad in Entrepreneurship** (Technology Track)
  - Finalist, 4th place out of 100 in team standings
- **VII Interuniversity Mathematical Battles Tournament, MIEM HSE**
  - Winner (team standings among first-year students)

Project Experience

Video Threat Detection Platform February – April 2025
Project, Final of All-Russian School Olympiad in Entrepreneurship

- **Task:** develop an applied solution for detecting threats in a video stream and displaying results.
- **My role:** end-to-end implementation: set the recognition quality requirements, fine-tuned a YOLOv8 model and built the application that processes the video stream in real time.
- **Tools:** Python, PyTorch, YOLOv8, CV, web interface.
- **Result:** implemented a working prototype, presented at the final of the All-Russian School Olympiad in Entrepreneurship; model quality improved relative to the baseline version.

Dialogue AI Companion with User Profile (NeuroLady) November 2025 – present
Personal Project

- **Task:** build robust dialogue handling logic accounting for history and user state.
- **My role:** designed the DB schema for model state storage, content generation logic considering user stories, processing modules and state update rules; implemented backend, unit tests, and background processes.
- **Tools:** Python, SQLite, SQLAlchemy, REST API, task scheduler.
- **Result:** covered 78% of critical API endpoints with tests, identified and fixed 12 edge-case data processing scenarios; after SQL query optimization and caching, average response latency dropped from 340 ms to 190 ms.

Training Dataset Preparation (startup Data2Use) March 2026 – present
Academic & Development Project

- **Task:** automate the formation of training samples and dataset creation from unstructured corporate data prior to use in LLM models.
- **My role:** described pipeline stages, implemented validation and preprocessing, added unit tests for key transformations; currently implementing mechanics for generating training samples from processed text blocks.
- **Tools:** Python, SQL, data processing scripts.
- **Result:** automated data quality analytics collection (duplicates, missing values, schema errors); parsing error count reduced by 65%.